YOLO
MobileNet

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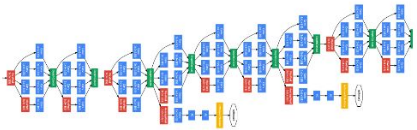
MSAI

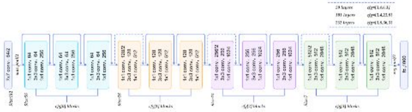
Recap

- Computer Vision tasks
  - Classification
  - Localisation
  - Detection
  - Semantic Segmentation
  - Instance Segmentation
- Metrics
  - IoU
  - mAP
- Datasets
  - PASCAL VOC
  - ImageNet
  - COCO
  - Open Images
- R-CNN
- Fast R-CNN
- Faster R-CNN

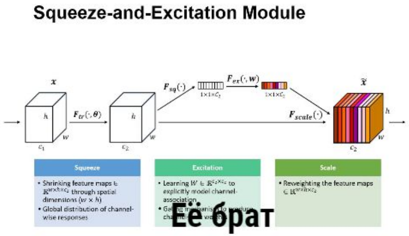
Девушка, которая мне нравится



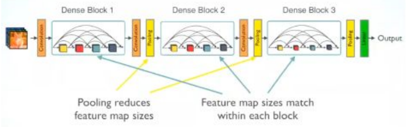
Её отец



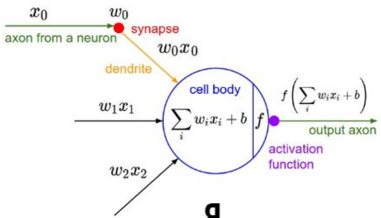
Её мать



Её брат



Её бывший



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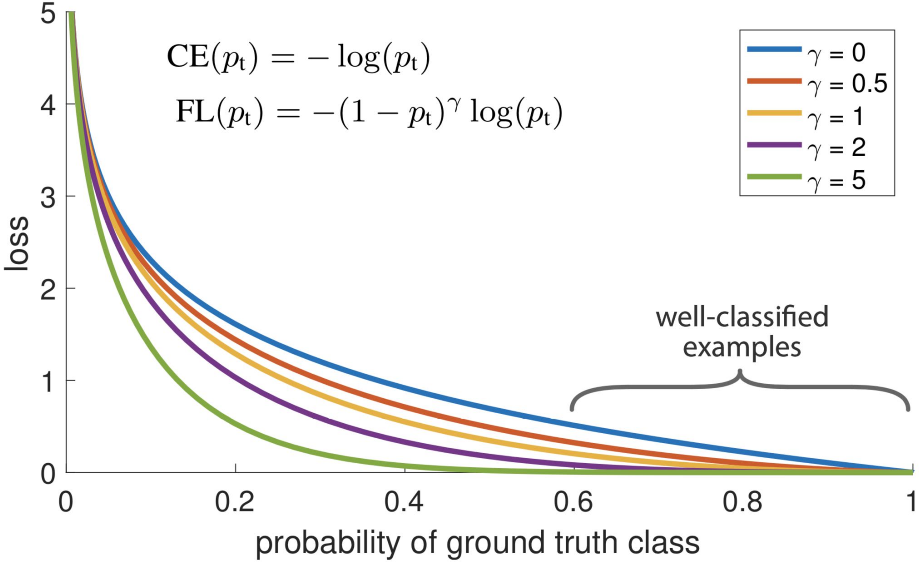
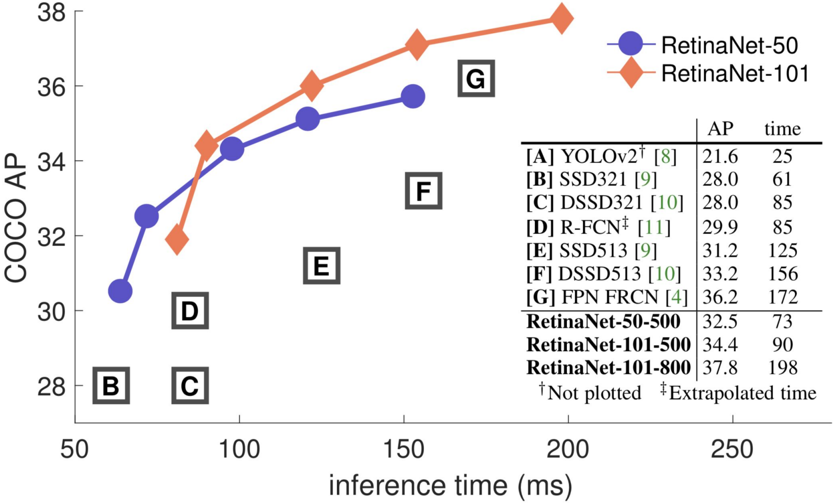
Outline

- Focal Loss
- Non-Maximum Suppression
- YOLO
    - v1
    - v2
    - V3
- Separable convolutions
- MobileNet

Focal Loss



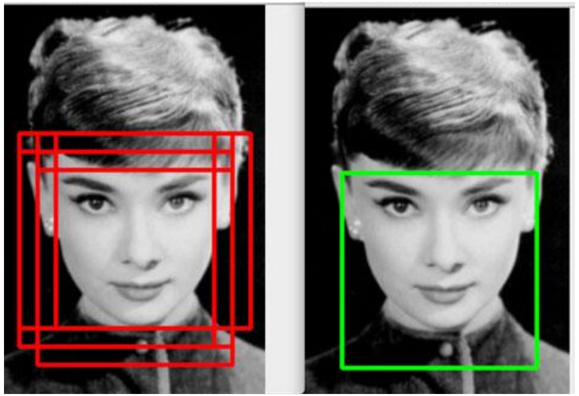


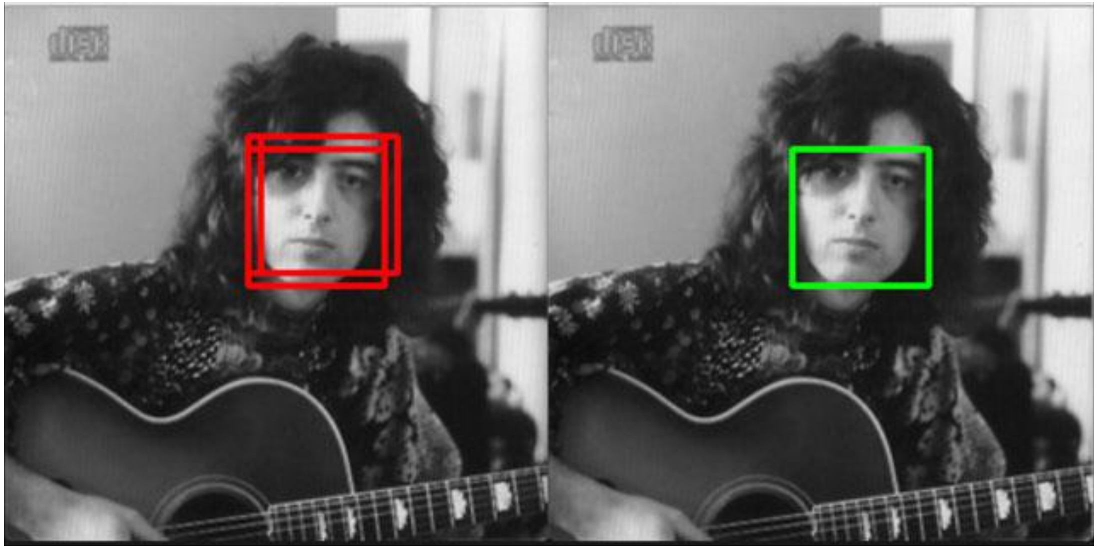
	AP	time
[A] YOLOv2† [8]	21.6	25
[B] SSD321 [9]	28.0	61
[C] DSSD321 [10]	28.0	85
[D] R-FCN‡ [11]	29.9	85
[E] SSD513 [9]	31.2	125
[F] DSSD513 [10]	33.2	156
[G] FPN FRCN [4]	36.2	172
RetinaNet-50-500	32.5	73
RetinaNet-101-500	34.4	90
RetinaNet-101-800	37.8	198

†Not plotted ‡Extrapolated time

[Video talk](#)

Non-Maximum Suppression





You Only Look Once (YOLO)

YOLOv1-v2

Original Joseph Redmon slides from conferences

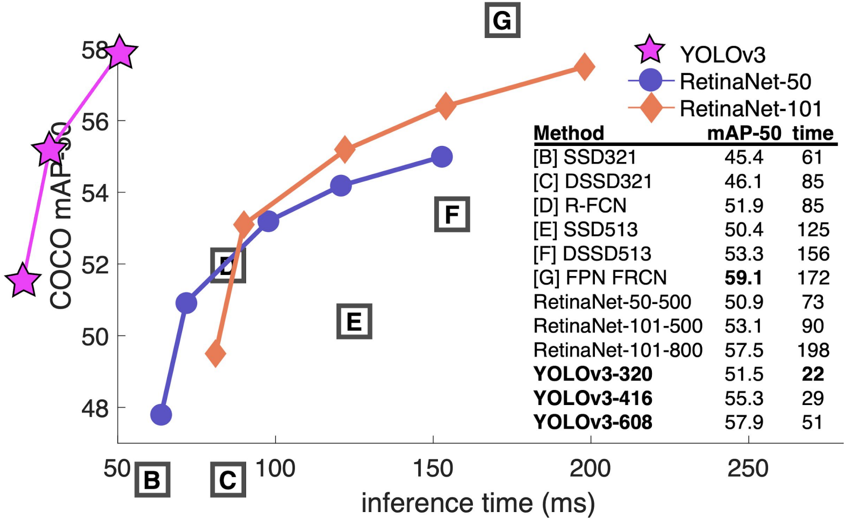
- YOLOv1
- YOLOv2

Also CVPR 2016 talk video

YOLOv3

- Bounding Box Prediction
- Class Prediction
- Predictions Across Scales
- Feature Extractor

All articles published on [Joseph's website]

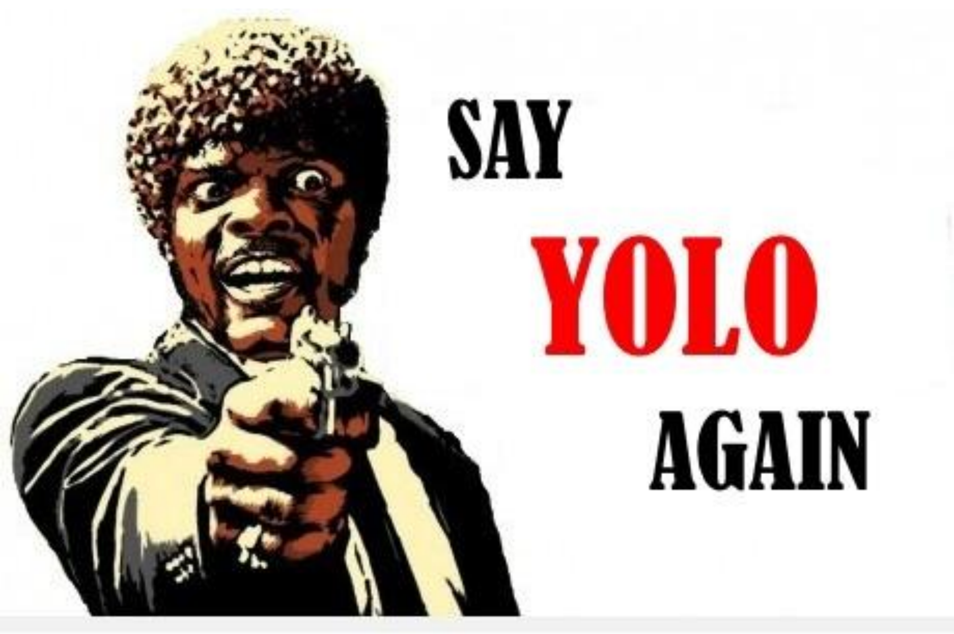


Practical notes

Write your own YOLO from scratch
https://blog.paperspace.com/how-to-implement-a-yolo-object-detector-in-pytorch/

Original YOLO written on Darknet - custom NN framework

Weights import to TF via
https://github.com/thtrieu/darkflow



Current state

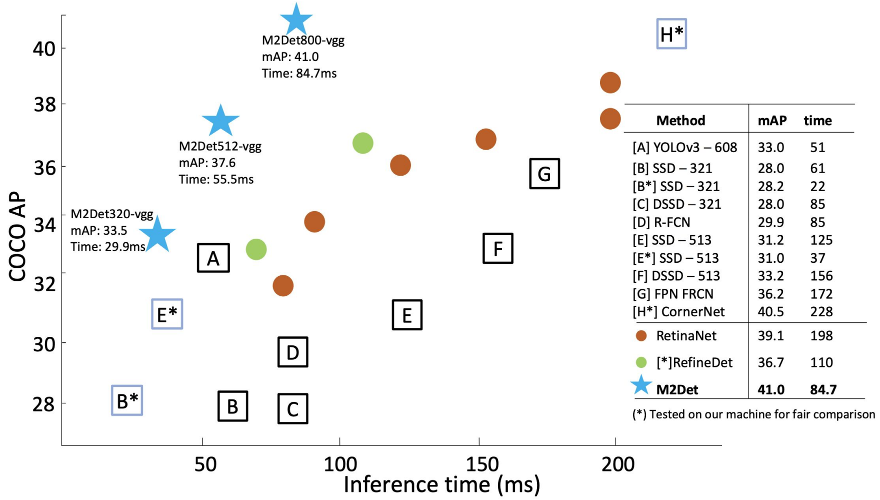


Figure 5: Speed (ms) vs. accuracy (mAP) on COCO *test-dev*.

M2Det: A Single-Shot Object Detector based on Multi-Level Feature Pyramid Network

https://arxiv.org/pdf/1811.04533.pdf

Who is responsible for all these YOLOs

https://docs.google.com/spreadsheets/d/1BxI0SkyFGNvfzLLM_46xZ3i5crTst7jsUKKuZZDWrA0/edit#gid=0

Tools for detection

https://github.com/roboflow/supervision

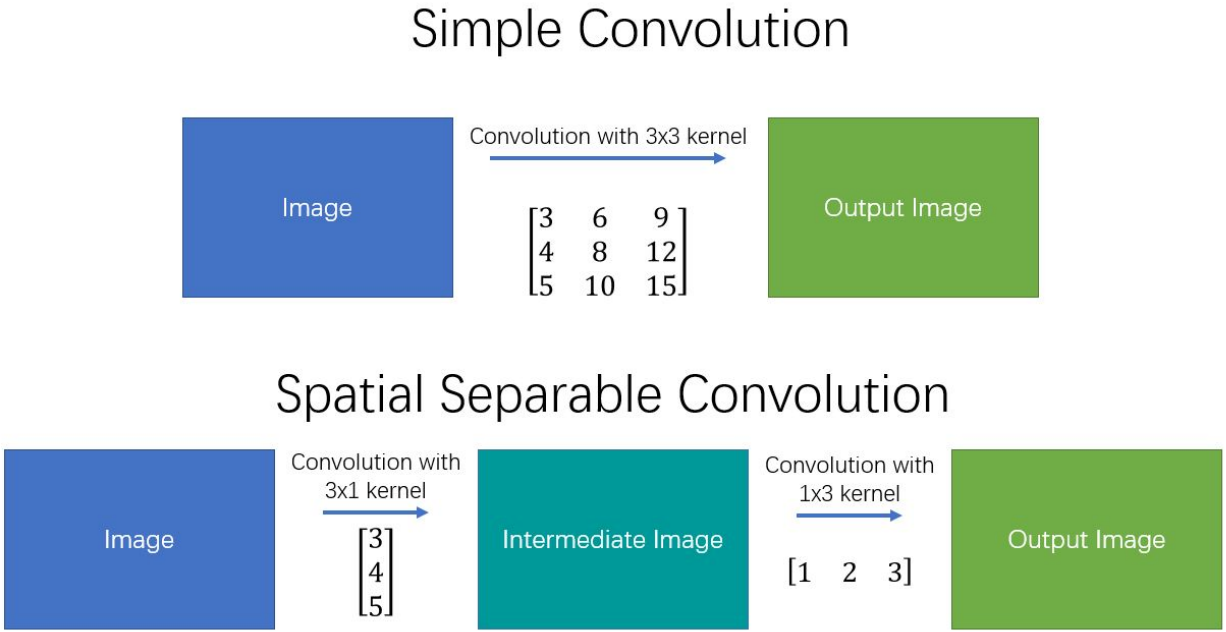
Separable convs
MobileNet

Spatial Separable Convolution

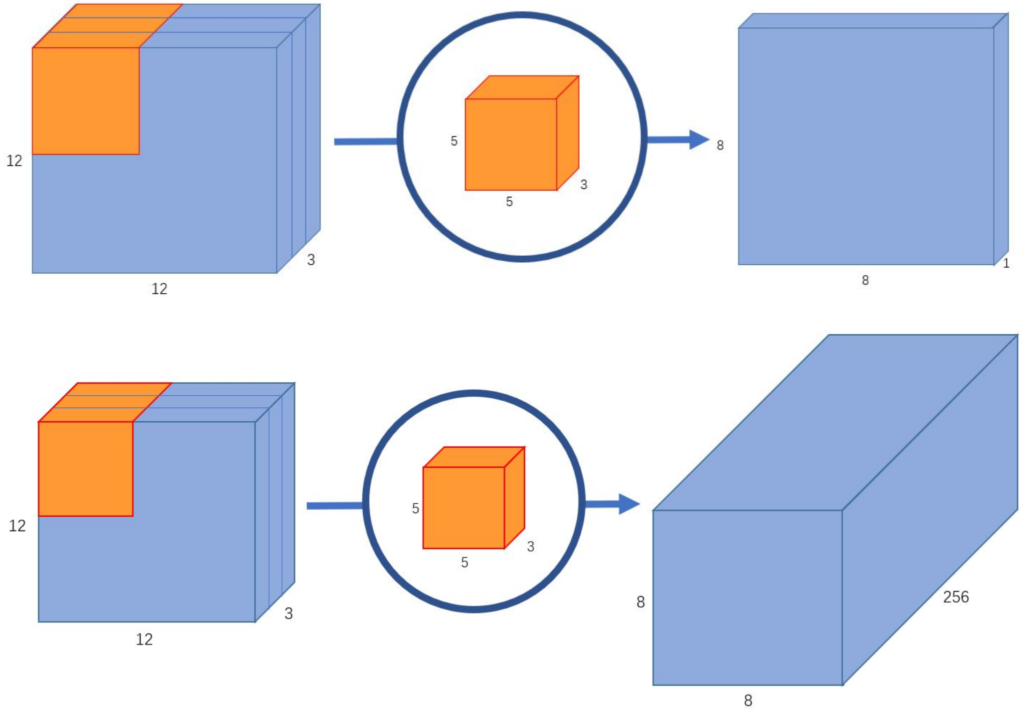
$$\begin{bmatrix} 3 & 6 & 9 \\ 4 & 8 & 12 \\ 5 & 10 & 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

[Images credentials]

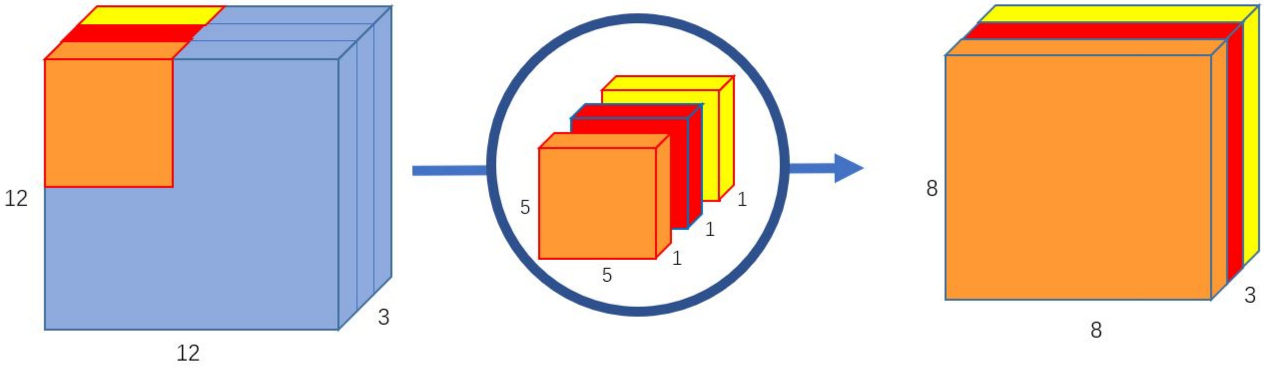
Spatial Separable Convolution



Normal Convolution

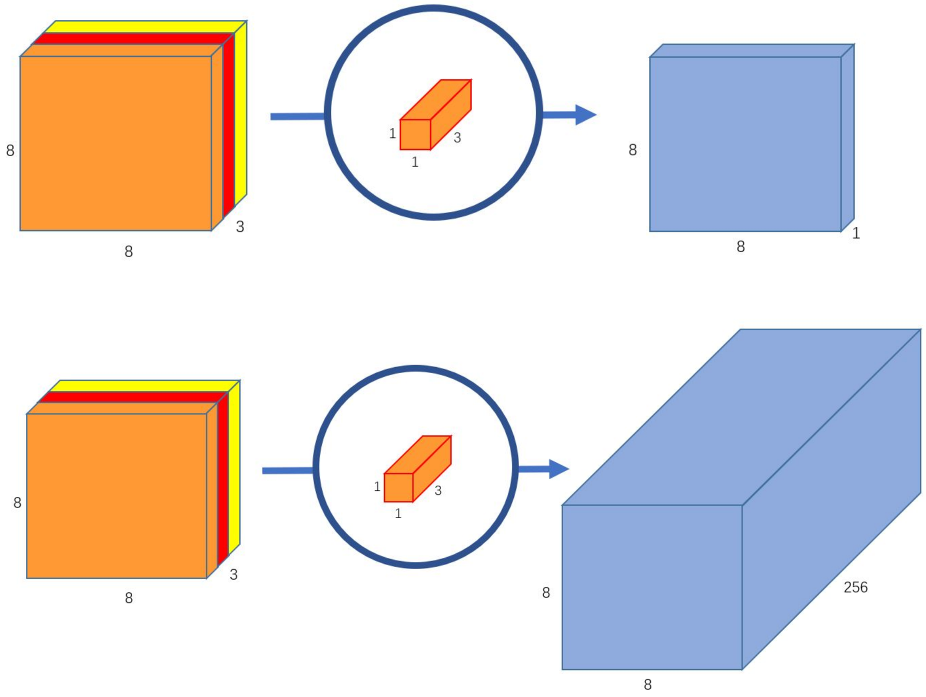


Depthwise Convolution



Don’t change channels number

Pointwise Convolution



MobileNet v1

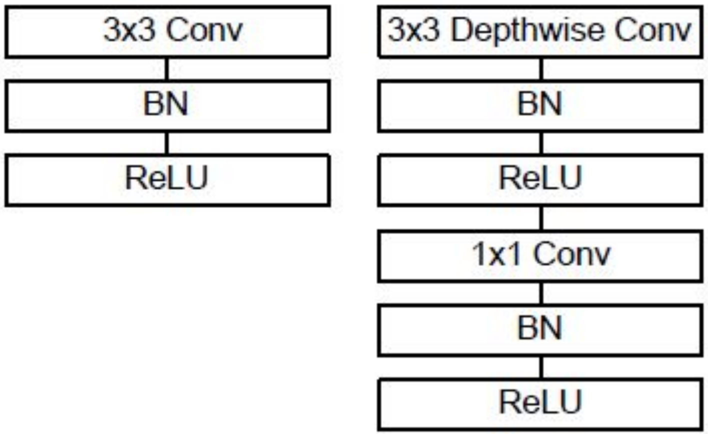
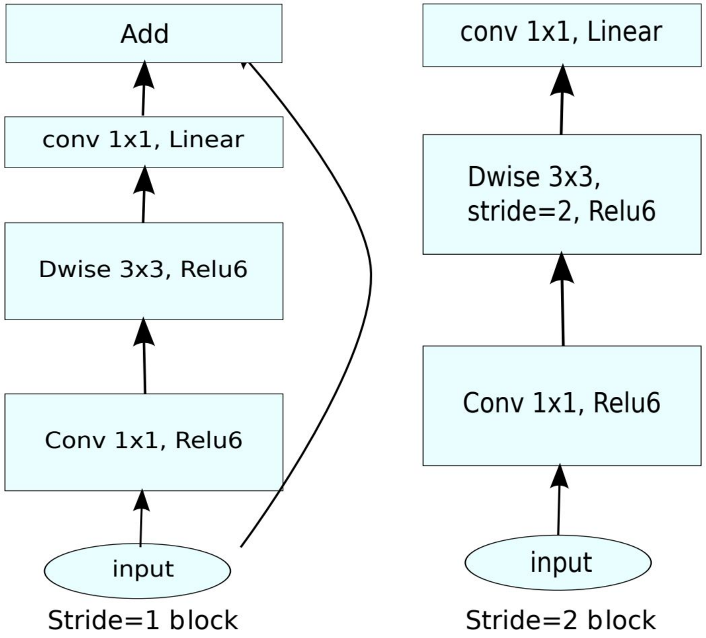
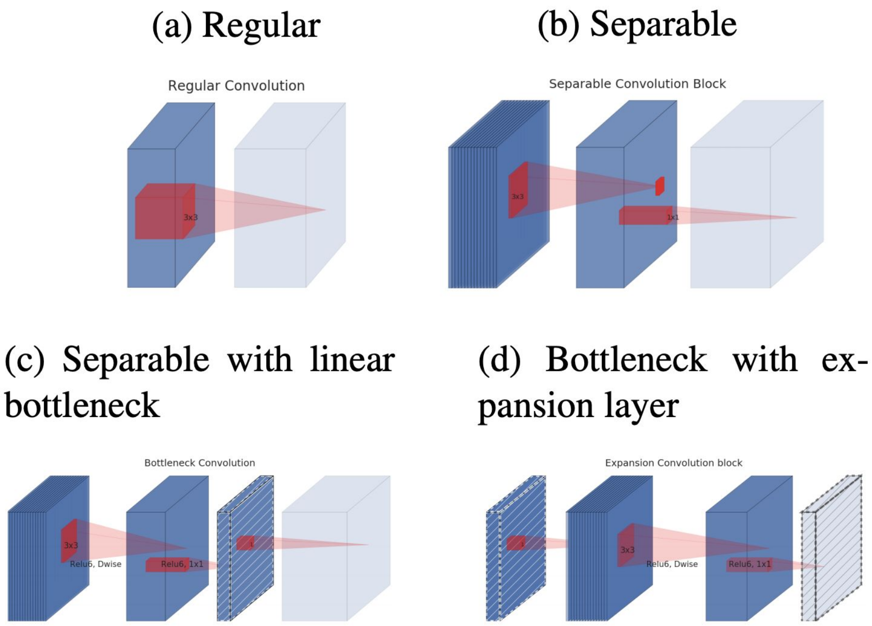


Table 1. MobileNet Body Architecture

Type / Stride	Filter Shape	Input Size
Conv / s2	$3 \times 3 \times 3 \times 32$	$224 \times 224 \times 3$
Conv dw / s1	$3 \times 3 \times 32$ dw	$112 \times 112 \times 32$
Conv / s1	$1 \times 1 \times 32 \times 64$	$112 \times 112 \times 32$
Conv dw / s2	$3 \times 3 \times 64$ dw	$112 \times 112 \times 64$
Conv / s1	$1 \times 1 \times 64 \times 128$	$56 \times 56 \times 64$
Conv dw / s1	$3 \times 3 \times 128$ dw	$56 \times 56 \times 128$
Conv / s1	$1 \times 1 \times 128 \times 128$	$56 \times 56 \times 128$
Conv dw / s2	$3 \times 3 \times 128$ dw	$56 \times 56 \times 128$
Conv / s1	$1 \times 1 \times 128 \times 256$	$28 \times 28 \times 128$
Conv dw / s1	$3 \times 3 \times 256$ dw	$28 \times 28 \times 256$
Conv / s1	$1 \times 1 \times 256 \times 256$	$28 \times 28 \times 256$
Conv dw / s2	$3 \times 3 \times 256$ dw	$28 \times 28 \times 256$
Conv / s1	$1 \times 1 \times 256 \times 512$	$14 \times 14 \times 256$
$5\times$ Conv dw / s1	$3 \times 3 \times 512$ dw	$14 \times 14 \times 512$
$5\times$ Conv / s1	$1 \times 1 \times 512 \times 512$	$14 \times 14 \times 512$
Conv dw / s2	$3 \times 3 \times 512$ dw	$14 \times 14 \times 512$
Conv / s1	$1 \times 1 \times 512 \times 1024$	$7 \times 7 \times 512$
Conv dw / s2	$3 \times 3 \times 1024$ dw	$7 \times 7 \times 1024$
Conv / s1	$1 \times 1 \times 1024 \times 1024$	$7 \times 7 \times 1024$
Avg Pool / s1	Pool 7×7	$7 \times 7 \times 1024$
FC / s1	1024×1000	$1 \times 1 \times 1024$
Softmax / s1	Classifier	$1 \times 1 \times 1000$

MobileNet v2





Results

Architecture	Params	**Top-1 accuracy**	**Top-5 accuracy**
Xception	22.91M	0.790	0.945
VGG16	138.35M	0.715	0.901
MobileNetV1 (alpha=1, rho=1)	4.20M	0.709	0.899
MobileNetV1 (alpha=0.75, rho=0.85)	2.59M	0.672	0.873
MobileNetV1 (alpha=0.25, rho=0.57)	0.47M	0.415	0.663
MobileNetV2 (alpha=1.4, rho=1)	6.06M	**0.750**	**0.925**
MobileNetV2 (alpha=1, rho=1)	3.47M	0.718	0.910
MobileNetV2 (alpha=0.35, rho=0.43)	1.66M	0.455	0.704

MobileNet v3

Layer removal

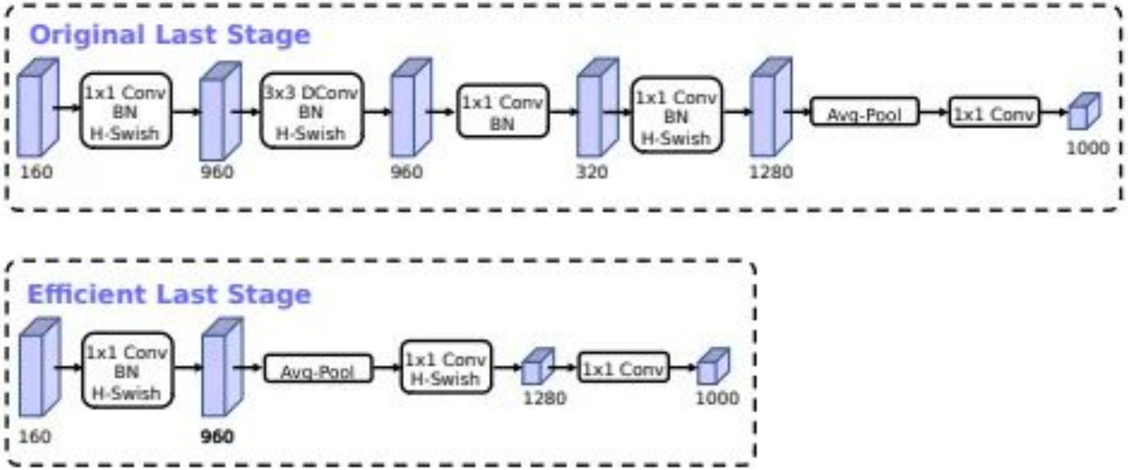


Figure 5. Comparison of original last stage and efficient last stage. This more efficient last stage is able to drop three expensive layers at the end of the network at no loss of accuracy.

MobileNet v3

Nonlinearity

$$\text{swish}\, x = x \cdot \sigma(x)$$

$$\text{h-swish}[x] = x\frac{\text{ReLU6}(x+3)}{6}$$

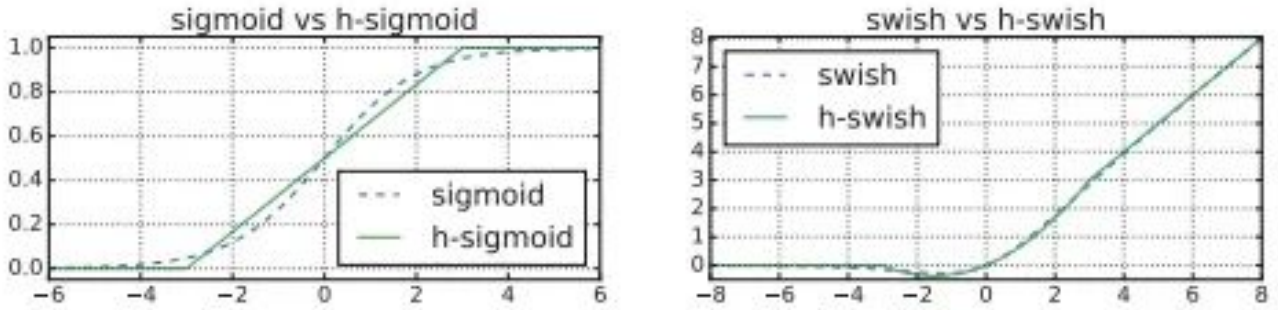


Figure 6. Sigmoid and swish nonlinearities and ther "hard" counterparts.

Results

Network	Top-1	MAdds	Params	P-1	P-2	P-3
V3-Large 1.0	**75.2**	**219**	5.4M	**51**	**61**	**44**
V3-Large 0.75	73.3	155	4.0M	39	46	40
MnasNet-A1	75.2	315	3.9M	71	86	61
Proxyless[5]	74.6	320	4.0M	72	84	60
V2 1.0	72.0	300	3.4M	64	76	56
V3-Small 1.0	**67.4**	**56**	2.5M	**15.8**	**19.4**	**14.4**
V3-Small 0.75	65.4	44	2.0M	12.8	15.6	11.7
Mnas-small [43]	64.9	65.1	1.9M	20.3	24.2	17.2
V2 0.35	60.8	59.2	1.6M	16.6	19.6	13.9

Table 3. Floating point performance on the Pixel family of phones (P-n denotes a Pixel-n phone). All latencies are in ms and are measured using a single large core with a batch size of one. Top-1 accuracy is on ImageNet.

Revise

- Recap
- Focal Loss
- Non-Maximum Suppression
- YOLO
- Separable convolutions
- MobileNet

Next time

- Segmentation

See Mask R-CNN ICCV17 talk to prepare yourself